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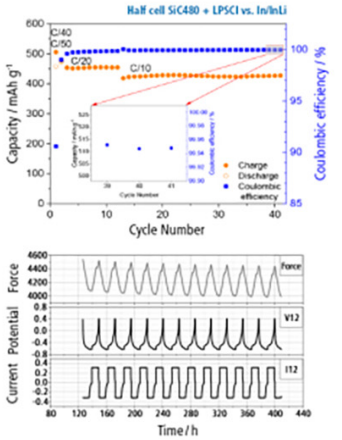
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A Database for Comparative Electrochemical Performance of Commercial 18650-Format Lithium-Ion Cells

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Lithium-ion batteries are a central technology to our daily lives with widespread use in mobile devices and electric vehicles. These batteries are also beginning to be widely used in electric grid infrastructure support applications which have stringent safety and reliability requirements. Typically, electrochemical performance data is not available for modelers to validate their simulations, mechanisms, and algorithms for lithium-ion battery performance and lifetime. In this paper, we report on the electrochemical performance of commercial 18650 cells at a variety of temperatures and discharge currents. We found that $LiFePO_4$ is temperature tolerant for discharge currents at or below 10 A whereas $LiCoO_2$, $LiNi_xCo_yAl_{1-x-y}O_2$, and $LiNi_{0.80}Mn_{0.15}Co_{0.05}O_2$ exhibited optimal electrochemical performance when the temperature is maintained at 15°C. $LiNi_xCo_yAl_{1-x-y}O_2$ showed signs of lithium plating at lower temperatures, evidenced by irreversible capacity loss and emergence of a high-voltage differential capacity peak. Furthermore, all cells need to be monitored for self-heating, as environment temperature and high discharge currents may elicit an unintended abuse condition. Overall, this study shows that lithium-ion batteries are highly application-specific and electrochemical behavior must be well understood for safe and reliable operation. Additionally, data collected in this study is available for anyone to download for further analysis and model validation.

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Lithium-ion batteries (LiBs) have become synonymous with portable electronics. The use of LiBs is expanding well beyond portable electronics into applications that require large energy and power capacities including electric vehicles,[1,2] and grid energy storage systems.[3,4] However, with increasing system size and large scale deployments, safety issues associated with large LiB system failures are becoming important topics.[5–8] When determining the safety and reliability of an energy storage system (ESS), system performance and capacity loss are predictive parameters. For the successful implementation and commercial operation of utility class energy storage plants, safety and reliability define commercial feasibility and success. Despite a substantial body of research detailing the development of LiB technology,[9–19] detailed studies on the electrochemical performance as a function of application conditions have been limited.[20–32]

When constructing an energy storage system, cells must be selected based on application-specific operation and performance characteristics. Application-specific cell-level requirements are highly unique and include operation temperatures and currents, stable output capacity and voltage, cell cycle or calendar life, and safety criteria. In support of this, cell manufacturers offer a wide variety of commercial cells that are designed to offer specific power and energy characteristics. Unfortunately, comparable information on electrochemical performance from cells made by different manufacturers is difficult to ascertain. While manufacturers do provide information ranging from complete data sheets with voltage profiles at various temperatures and discharge currents, these specifications often lack fidelity and parametric detail to ascertain the reliability and operational performance. In addition, as manufacturer specifications are based on proprietary testing, replicating cell performance by an energy storage system integrator is a challenging task.

When choosing cells to build an application-specific energy storage system, there are a number of parameters and performance characteristics that must be carefully considered. Initially, physical size, shape, and weight must conform to a battery pack specification. Among the most common cells are the 18650 format cylindrical cells, with a diameter of 18 mm and 65 mm long. Though 18650 cells were originally designed for portable electronics, these cells are now widely used in power tools, electric vehicle batteries, and stationary energy storage applications.[33,34] Due to the wide ranging use of 18650s across a variety of applications, we decided to use this cell format for the current study.

In most cases, general electrochemical performance information can be found in the manufacturer data sheets; including allowable operating temperature, maximum charging and discharging currents, and nominal cell capacity. However, this data often lacks details on the operational performance under a variety of conditions within the specified operating ranges. For example, output capacity as a function of discharge rate and environment temperature is critical information when choosing the optimal cell for a specific application. Additionally, one must carefully consider compounding operating conditions such as discharge current and environment temperature which may interact and alter the overall cell performance, reliability, and cell safety. The number of variables involved and the range of these parameters to be considered including cell chemistry, capacity, manufacturer, size, cycling profile, and environment temperature make it difficult to compare performance across a range of cell chemistries and manufacturers.

In this paper, we studied the electrochemical performance of four commercially relevant cell chemistries made by three reputable manufacturers: A123 (http://www.a123systems.com/lifepo4-battery-cell.htm), LG Chem (http://lg18650.com/2_lg-chem), and Panasonic (http://www.panasonic-batteries.com/). We measured the output capacity as a function of discharge current while monitoring the cells' external temperature. In addition, we used electrochemical impedance spectroscopy (EIS) techniques and made differential capacity measurements to understand the reversible and irreversible changes from various operating conditions. We also made an effort to quantify compounding effects of overlapping operating conditions with an eye on safety.

Testing Methodology

Experimental set-up.—Cycling studies were conducted using an Arbin SCTS multi-channel battery testing system. Battery cells were connected with commercially available 18650 battery holders. Cell holders were connected to the Arbin using appropriately rated wiring for current. Length of cabling was less than two meters to limit voltage drop. Cells being cycled were placed in an SPX Tenney Model T10C-1.5 environmental chamber, where environment temperatures could be controlled between −73°C and 200°C. Cell temperature was monitored with a thermocouple attached to the skin of each cell

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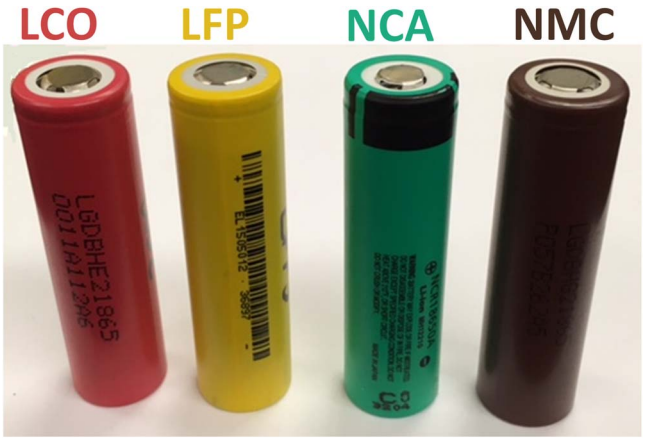


Figure 1. Cells tested in this work. From left to right: LGDBHE21865 (LCO), ANR18650M1 (LFP), NCR18650A (NCA), and LGDBHG21865 (NMC).

under test. Electrochemical impedance spectroscopy was conducted using a Solartron SI 1287 Electrochemical Interface and a SI 1260 Impedance/Gain-Phase Analyzer. For these measurements, cells were connected in the same configuration as was used for cycling studies.

The four cells used in this work corresponding to four cell chemistries, namely $LiCoO_2$ (LCO, Part # LGDBHE21865, made by LG Chem), $LiFePO_4$ (LFP, Part # ANR18650M1, made by A123), $LiNi_xCo_yAl_{1-x-y}O_2$ (NCA, Part # NCR18650A, made by Panasonic), and $LiNi_{0.80}Mn_{0.15}Co_{0.05}$ (NMC, Part # LGDBHG21865, made by LG Chem) are shown in Figure 1. Available information on manufacturer specified operating conditions along with electrode composition results and some key electrochemical performance parameters is summarized in Table I. However, detailed information regarding electrolyte and separator compositions was not obtainable. Based on the available industry data, it is likely that electrolytes are carbonate-based with a lithium salt.[35] To ensure that the cells are genuine, they were purchased from reputable manufacturer's representatives. Unless otherwise stated, manufacturer suggested operating conditions are used as the safe operating window. In other words, the maximum temperature was used as a test abort condition if registered by a thermocouple attached to the cell's skin.

Testing procedures.—An electrochemical testing procedure for each cell consisted of six stages as shown in Figure 2. In the first stage, three examples of each cell were discharged to 0% state of charge (SOC) to ensure consistent EIS results. Cells were then placed in an environmental chamber at the cycling temperature and equilibrated for over 12 hours. Five temperatures were probed in this work, consisting of 5, 15, 25, 35, and 45°C. EIS was performed over a frequency range of 0.1 to 100,000 Hz with a 0.010 V perturbation. Cells were also cycled through various discharge currents as shown in Figure 3. Cycling performance was performed twice, once as detailed

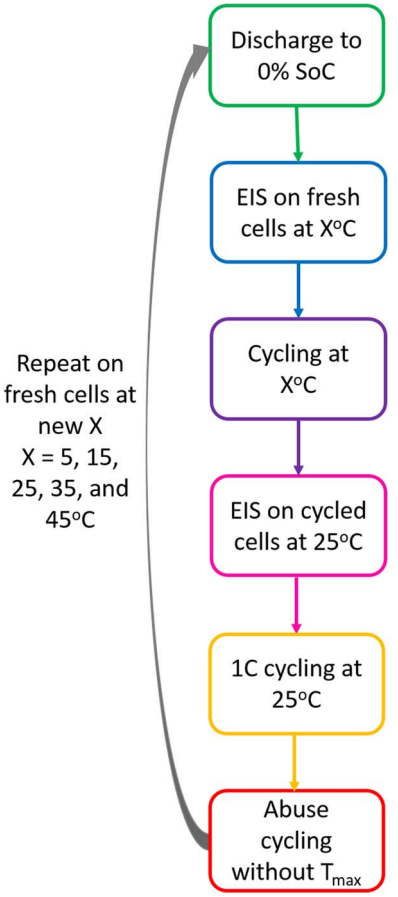


Figure 2. Schematic representation of testing procedures used in this study.

Table I. Commercial 18650-format lithium-ion battery information including electrode composition and manufacturer specified operating conditions.

Battery	LCO	LFP	NCA	NMC
Anode	Graphitic carbon	Nano-carbon	Graphitic carbon	Graphitic Carbon
Cathode	$LiCoO_2$	$LiFePO_4$	$LiNi_xCo_yAl_{1-x-y}O_2$	$LiNi_{0.80}Mn_{0.15}Co_{0.05}O_2$
Nominal Capacity (Ah)	2.5	1.1	2.9	3.0
Nominal Voltage (V)	3.6	3.3	3.6	3.6
Measured Capacity[a] (Ah)	2.44 ± 0.03	1.062 ± 0.006	2.6 ± 0.2	2.8 ± 0.1
Measured Energy Density[a] (Wh)	8.8 ± 0.1	3.50 ± 0.02	9.4 ± 0.6	10.1 ± 0.4
Maximum Discharge Current (A)	20	30	6	20
Operating Temperature Range (°C)	0 to 50	−30 to 60	0 to 45	0 to 50
Nominal Volume (mL)	16.5	16.5	16.5	16.5
Measured Mass[b] (g)	44.9	39.4	44.8	44.9
Cost per Cell[c] ($)	7.50	9.95	7.85	6.00
Energy Density (Wh/L)	533.3	212.1	569.7	612.1
Specific Energy (Wh/kg)	195.8	88.9	210.0	224.8
Cost per Capacity ($/kWh)	852.3	2,842.8	835.1	594.0

[a]measured from the second discharge cycle under 1C charge and discharge rates at 25°C.
[b]mass of 18650 without plastic label (jellyroll in 18650 can).
[c]cost from batteryspace.com (LCO, LFP, and NMC) and imrbatteries.com (NCA) June 2017.

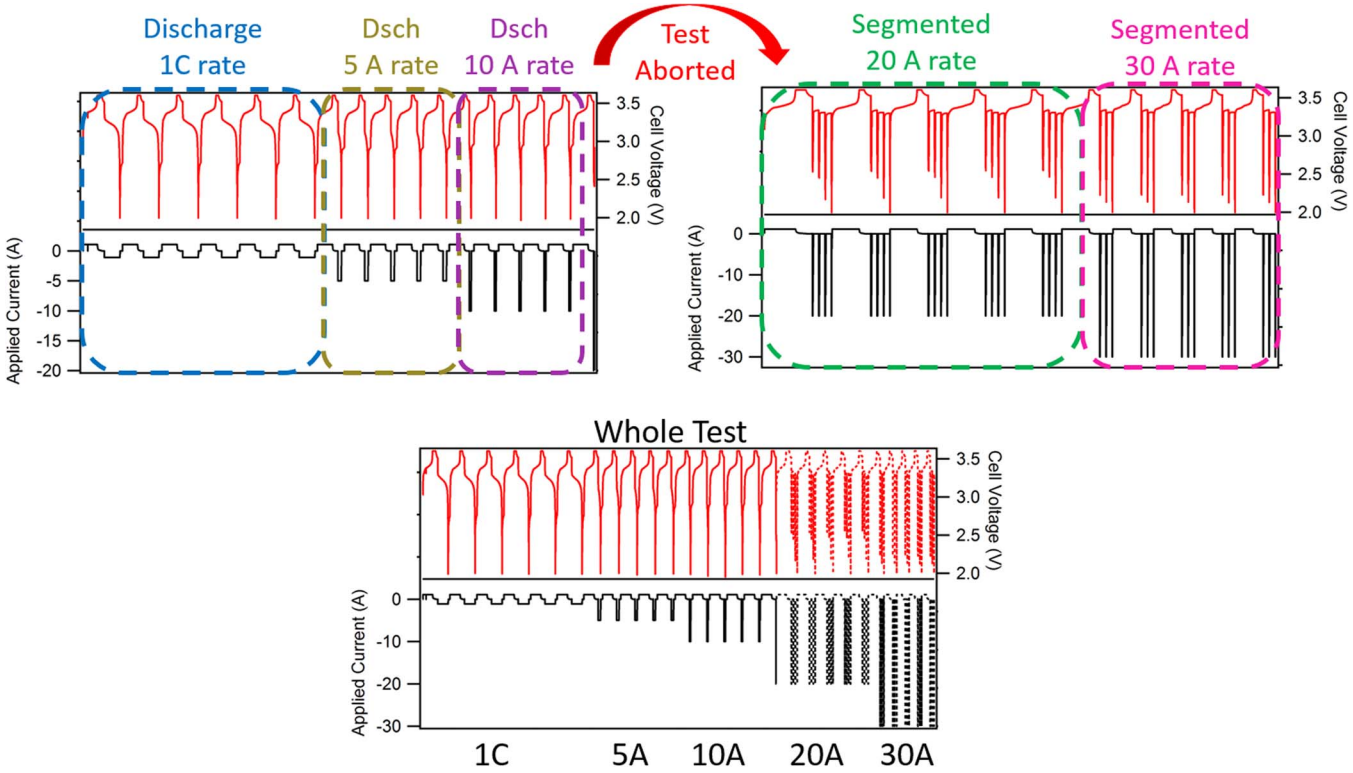


Figure 3. Electrochemical testing procedure for the LFP cell at 25°C. When presented in this work, the discharge capacity will be indicated at the top of the graph. If the cell's skin temperature reaches the maximum allowed, testing will abort. If aborted, an alternate test scheme employing a segmented discharge allows the test to run to completion. Segmented discharging includes additional rest steps and therefore is indicated with a dashed line.

here on a set of three cells and a second time without corresponding EIS before and after testing. This gives a total of six cells which were used to find the average and the standard deviation to prepare Figures 4 and 5. During cycling, cells were charged using a 1C charge rate followed by constant voltage until the current fell to 50 mA. Between each charge and discharge, the cells were rested for 10 minutes. To monitor cell temperature during cycling, a thermocouple was attached to the surface of each cell. This thermocouple measured the external cell temperature. Electrochemical tests were discontinued if the maximum manufacturer specified operating temperature was reached. Cells were cycled from 100% SOC to 0% SOC initially at 1C charging and discharging rates. After six cycles, the discharge rate was stepped up to include five cycles at a higher current (i.e., 5A, 10A, etc). Throughout testing, charging was maintained at a 1C rate. Increasing discharge current cycling was continued, with five cycles per discharge current, until the manufacturer specified maximum was reached. If the cell self-heated to the maximum temperature, testing was immediately discontinued. To complete the test, remaining discharge steps were segmented with additional rest steps. Including more rest steps allows for cooling between discharges and limits self-heating. Segmented discharge data is indicated in figures with a dashed line as cell behavior might be altered by inclusion of the additional rests. EIS was recorded on each cell once again at room temperature. To test for lasting performance decay, a baseline 1C charge/discharge cycling was re-run at 25°C and plotted as the final cycles in figures to follow. This entire cycle was repeated for each temperature.

As a continuation of electrochemical testing described above, cells were tested under operating conditions beyond the manufacturer specifications, these tests are typically referred to as abuse testing. Cycling was continued from the abort step described above without a temperature cutoff. Pristine, uncycled cells were used for abuse testing to avoid previous capacity loss carry-over. Three examples of each cell were placed in an environmental chamber set at the desired temperature. A thermocouple was attached to the skin of each cell. No temperature abort conditions were set up in the Arbin software. These tests are designed to monitor cell electrochemical performance as well as maximum skin temperature.

Results and Discussion

Data repository.—Raw data from cycling, abuse, and EIS is published online through the ESS Safety Working Group data repository. This repository is free of charge and can be found on the web at http://www.sandia.gov/energystoragesafety/research-development/research-data-repository/. The data published online is the raw, unprocessed form used to prepare all electrochemical data, tables, and figures within this manuscript. The goal of having open data for this and other related works by these authors and others working in this area is to improve the integrity of publication findings, encourage collaborative research, enable larger studies in the future that may leverage multiple sources of data from different studies, and encourage modeling efforts by other researchers who do not otherwise have access to experimental data.

Initial testing.—Cells are often compared in terms of energy density, specific energy, and cost per capacity. Measured capacity and measured energy density were established from the second discharge at 1C charge and discharge rates and 25°C in six examples of each cell. Using mass, volume, and cost per cell; energy density, specific energy, and cost per capacity were calculated and shown in Table I. All cells achieved capacities near their listed nominal capacity. Minor deviations in initial capacity occurred in NCA and NMC cells, which also had the largest deviation from the manufacturer's specification. NCA and NMC cells have a measured capacity near 3 Ah. LCO has a measured capacity around 2.5 Ah. LFP has a low measured capacity of 1.1 Ah but a similar mass, volume, and price to the LCO, NCA, and NMC cells. However, focusing on a single performance parameter such as energy density, specific energy, or cost per capacity can

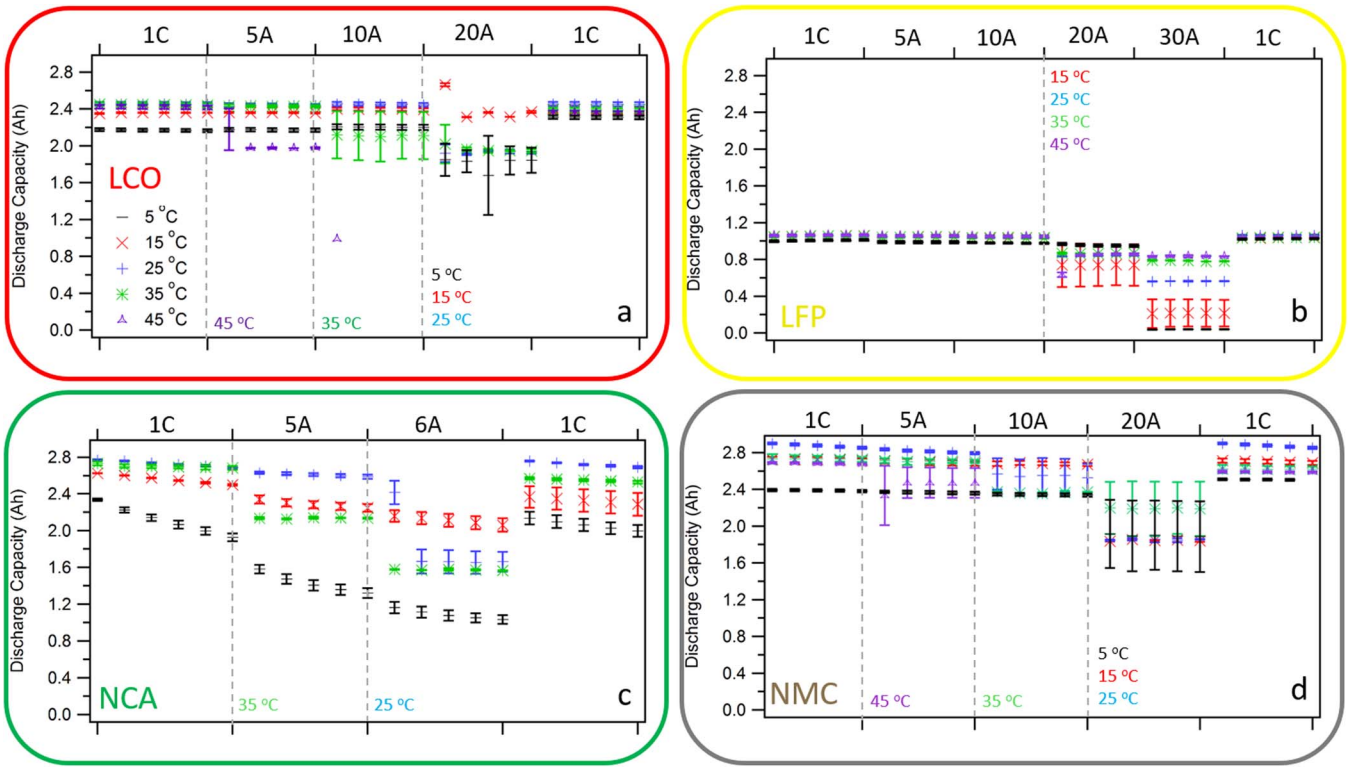


Figure 4. Six examples of each cell type were cycled, averaged together, and standard deviation was found to prepare this figure. The testing procedure shown in Figure 3 was repeated at 5°C (black dash), 15°C (red x), 25°C (blue cross), 35°C (green star), and 45°C (purple triangle). Discharge capacity as a function of discharge current (indicated on top of the graph, five cycles each) and temperature for each cell tested: (a) LCO (red outline, top-left), (b) LFP (yellow outline, top-right), (c) NCA (green outline, bottom-left), and (d) NMC (brown, bottom-right). Recall that the vertical dashed lines indicate the start of segmented discharging for the indicated temperature. Testing at 45°C on LCO and NMC was discontinued at higher currents due to self-heating and the cutoff of 50°C. The final 1C discharge rate was performed at 25°C after cycling. This information is included to demonstrate any lasting capacity loss.

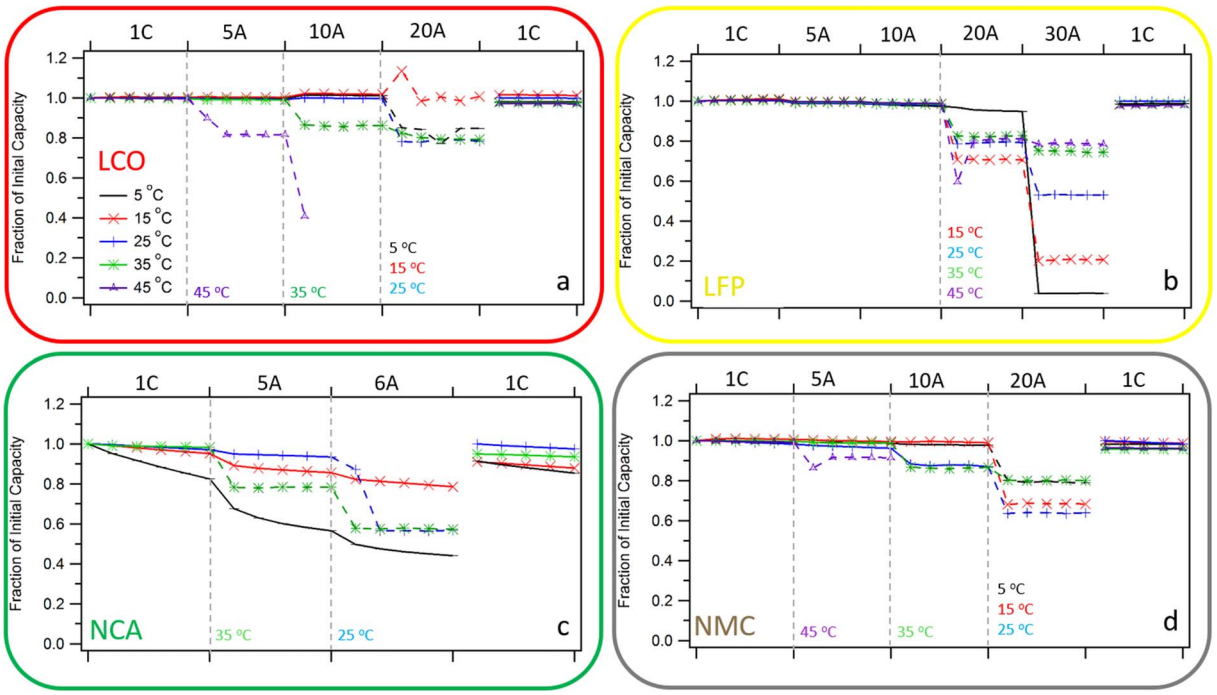


Figure 5. Six examples of each cell type were cycled and averaged together to prepare this figure. The testing procedure shown in Figure 3 was repeated at 5°C (black dash), 15°C (red x), 25°C (blue cross), 35°C (green star), and 45°C (purple triangle). The fraction of initial capacity as a function of discharge current (indicated on top of the graph, five cycles each) and temperature for each cell tested: (a) LCO (red outline, top-left), (b) LFP (yellow outline, top-right), (c) NCA (green outline, bottom-left), and (d) NMC (brown outline, bottom-right). Recall that the vertical dashed lines indicate the start of segmented discharging for the indicated temperature. Testing at 45°C on LCO and NMC was discontinued at higher currents due to self-heating and the cutoff of 50°C. The final 1C discharge rate was performed at 25°C after cycling. This information is included to demonstrate any lasting capacity loss.

Table II. Summary of discharge capacity under various operating conditions. Values are taken from the second cycle of the given conditions. Bold values are segmented discharge capacities. Blank values represent conditions outside the manufacturer specified operating window.

Test	Temp (°C)	Discharge Current (A)	LCO (Ah)	LFP (Ah)	NCA (Ah)	NMC (Ah)
Cycling (Figure 4)	5	1C‡	2.17 ± 0.02	1.00 ± 0.01	2.22 ± 0.03	2.39 ± 0.01
		5	2.17 ± 0.02	0.99 ± 0.01	1.47 ± 0.05	2.37 ± 0.02
		10†	2.20 ± 0.03	0.98 ± 0.01	1.11 ± 0.06	2.34 ± 0.02
		20	**1.8 ± 0.1**	0.97 ± 0.01	—	**1.9 ± 0.4**
		30	—	0.037 ± 0.002	—	—
		post- test 25	2.31 ± 0.02	1.023 ± 0.009	2.10 ± 0.07	2.51 ± 0.01
	15	1C‡	2.360 ± 0.004	1.050 ± 0.002	2.601 ± 0.003	2.72 ± 0.03
		5	2.360 ± 0.004	1.043 ± 0.004	2.30 ± 0.04	2.70 ± 0.03
		10†	2.40 ± 0.02	1.032 ± 0.004	2.14 ± 0.07	2.69 ± 0.02
		20	**2.312 ± 0.003**	**0.7 ± 0.2**	—	**1.855 ± 0.008**
		30	—	**0.2 ± 0.1**	—	—
		post- test 25	2.39 ± 0.04	1.033 ± 0.005	2.3 ± 0.1	2.71 ± 0.02
	25	1C‡	2.45 ± 0.02	1.061 ± 0.006	2.758 ± 0.006	2.90 ± 0.01
		5	2.44 ± 0.02	1.048 ± 0.004	2.62 ± 0.02	2.82 ± 0.01
		10†	2.45 ± 0.02	1.041 ± 0.004	**1.7 ± 0.1**	2.54 ± 0.02
		20	**1.91 ± 0.01**	**0.836 ± 0.002**	—	**1.86 ± 0.03**
		30	—	**0.564 ± 0.001**	—	—
		post- test 25	2.45 ± 0.02	1.062 ± 0.007	2.737 ± 0.007	2.89 ± 0.01
	35	1C‡	2.45 ± 0.02	1.052 ± 0.009	2.70 ± 0.02	2.73 ± 0.05
		5	2.43 ± 0.01	1.044 ± 0.009	**2.128 ± 0.005**	2.70 ± 0.02
		10†	**2.1 ± 0.3**	1.037 ± 0.009	**1.569 ± 0.002**	**2.71 ± 0.02**
		20	**1.96 ± 0.02**	**0.862 ± 0.006**	—	**2.2 ± 0.3**
		30	—	**0.788 ± 0.004**	—	—
		post- test 25	2.39 ± 0.02	1.037 ± 0.009	2.56 ± 0.01	2.62 ± 0.04
	45	1C‡	2.42 ± 0.02	1.065 ± 0.008	—	2.694 ± 0.004
		5	**1.977 ± 0.003**	1.060 ± 0.009	—	**2.5 ± 0.2**
		10	—	1.05 ± 0.01	—	—
		20	—	**0.851 ± 0.002**	—	—
		30	—	**0.840 ± 0.006**	—	—
		post- test 25	2.36 ± 0.02	1.042 ± 0.008	—	2.595 ± 0.006
Abuse	5	20	2.20 ± 0.03	—	—	2.33 ± 0.09
		30	—	—	—	—
		post- test 25	2.42 ± 0.01			2.72 ± 0.01
	15	20	2.24 ± 0.02	0.056 ± 0.004	—	2.5 ± 0.2
		30	—	0.020 ± 0.006	—	—
		post- test 25	2.42 ± 0.02	1.048 ± 0.004		2.72 ± 0.01
	25	10†	—	—	2.38 ± 0.05	—
		20	2.32 ± 0.09	0.21 ± 0.03	—	2.63 ± 0.08
		30	—	0.010 ± 0.005	—	—
		post- test 25	2.43 ± 0.03	1.040 ± 0.002	2.655 ± 0.006	2.77 ± 0.02
	35	5	—	—	2.70 ± 0.01	—
		10†	2.48 ± 0.02	—	2.59 ± 0.01	2.860 ± 0.002
		20	2.3 ± 0.1	0.34 ± 0.04	—	2.63 ± 0.07
		30	—	0.020 ± 0.003	—	—
		post- test 25	2.410 ± 0.008	1.05 ± 0.01	2.69 ± 0.01	2.717 ± 0.004
	45	5	2.48 ± 0.02	—	—	2.92 ± 0.02
		10	2.44 ± 0.02	—	—	2.84 ± 0.02
		20	1.7 ± 0.5	0.3 ± 0.2	—	2.61 ± 0.01
		30	—	0.02 ± 0.01	—	—
		post- test 25	2.37 ± 0.01	1.041 ± 0.008	—	2.68 ± 0.02

‡1C is 2.5 A for LCO cells, 1.1 A for LFP cells, 2.7 A for NCA cells, and 3.0 A for NMC cells.
†6 A for NCA cells.

be deceiving. For example, although outside the scope of this study, LFP is known to have a long cycle and calendar life,[36–38] enhancing commercial viability. Additionally, LFP has excellent thermal stability and abuse tolerance,[34,39] furthering commercial attractiveness. These diverse performance parameters illustrate the complicated process of choosing the optimal cell.

Cycling.—To assess the impact of operating conditions, six examples of each cell were subjected to charge/discharge cycling with varying environment temperature and discharge currents. All cells underwent full cycles, from 100% SOC to 0% SOC as defined by the manufacturer. If the cell skin reached the maximum operating temperature, tests were aborted. Testing was repeated at each environment temperature. Segmented data is indicated with a dashed line. Segmented data may display artificially low discharge capacity and maximum temperature as a result of increased rest steps. Results are plotted as discharge capacity versus cycle in Figure 4 and summarized in Table II. The final five cycles were obtained at 1C rates and 25°C environment temperature and are also included in Table II as 'post-test 25'. This data serves to check for irreversible capacity loss after cycling. The y-axis is maintained over a constant range for ease of comparison across cells. Testing at 45°C on LCO and NMC was

discontinued at higher currents due to self-heating and the cutoff of 50°C. Testing was not performed at 45°C on NCA cells as this was the maximum allowable temperature. It is quickly apparent that environment temperature and discharge current have a profound effect on most cells. However, from 1C post-test cycling most of these effects are reversible. LCO, NCA, and NMC responded similarly to 15, 25, and 35°C, with lower temperatures favored at higher discharge currents. Extreme temperatures typically caused capacity loss which was reversible for most conditions. NCA cells in Figure 4c experienced irreversible capacity loss, 5.4% at 5°C and 11.4% at 15°C. All other cells tested regained near initial capacity after cycling. LFP in Figure 4b was stable until 20 A discharge, after which 35 and 45°C have the greatest capacity. In the LFP system, a more profound temporary capacity loss was observed with temperatures at or below 25°C and currents above 10 A.

In a real system, the fraction of initial capacity represents impacts of capacity loss with operating conditions. The fraction of initial capacity was found by dividing all discharge capacities reported in Figure 4 by that temperature's second cycle discharge capacity (1C discharge rate). Fraction of initial capacity as a function of cycle and temperature are presented in Figure 5. LCO and NMC cells maintain over 75% of initial capacity over all temperatures and discharge currents. NCA was found to be sensitive to low temperatures, with 5°C conditions losing over 50% of initial capacity. LFP was found to be stable up to 10A, after which it experiences profound capacity losses at temperatures less than 35°C. Again, minimal irreversible capacity loss is noted when post-test cycled at 1C and 25°C environment temperature. Even NCA maintained over 80% initial capacity after cycling.

Extent of self-heating.—While operating within the manufacturer guidelines, abuse conditions can be elicited. Abuse conditions are defined as anything outside of what is detailed within the manufacturer data sheet. Abuse conditions can cause cells to age prematurely or may lead to cell venting or thermal runaway. Cell self-heating is pronounced at elevated temperatures and high discharge currents. This is a critical parameter to understand when choosing cells for demanding applications such as electric vehicles and grid energy storage. During discharge, lithium ions travel from the carbon anode through the electrolyte and separator, to the intercalation compound cathode. At high currents the internal cell resistance causes the discharge mechanism to generate heat. High environment temperatures reduce the allowable heating window during discharge, before tests are aborted. Lower temperatures act to increase the resistance of the electrolyte, thus reducing ion mobility. This is countered by the larger allowable heating window experienced by cells in cool environments. Ultimately, cell composition, design, and additives dictate a cell's vulnerability to self-heating.

To probe the extend of self-heating and impacts of overtemperature during operation on cell performance, abuse testing was performed. Discharge cycling was resumed on three examples of each cell without temperature safety limits at the abort step described above and shown in Figures 4 and 5. Data was collected in exactly the same way and no other alterations were made to the cycling test or experimental set-up. Resulting fraction of initial capacity is plotted in Figure 6 and discharge capacities are listed in Table II. Comparing Figure 5 and Figure 6, dashed data from segmented discharging was replaced with abuse test data. In Figure 6a, the fraction of initial capacity is maintained near one throughout various environment temperatures and discharge currents. LCO cells are therefore resilient to temperature and current when it comes to electrochemical performance. This phenomenon was also observed in Figure 6d with NMC cells. NCA cells also demonstrated better performance without segmented discharging as seen in Figure 6c and Table II. NCA lost less capacity without segmentation than was observed in Figure 5c. In Figure 6b, LFP cells were resilient until 20 A discharging rates and

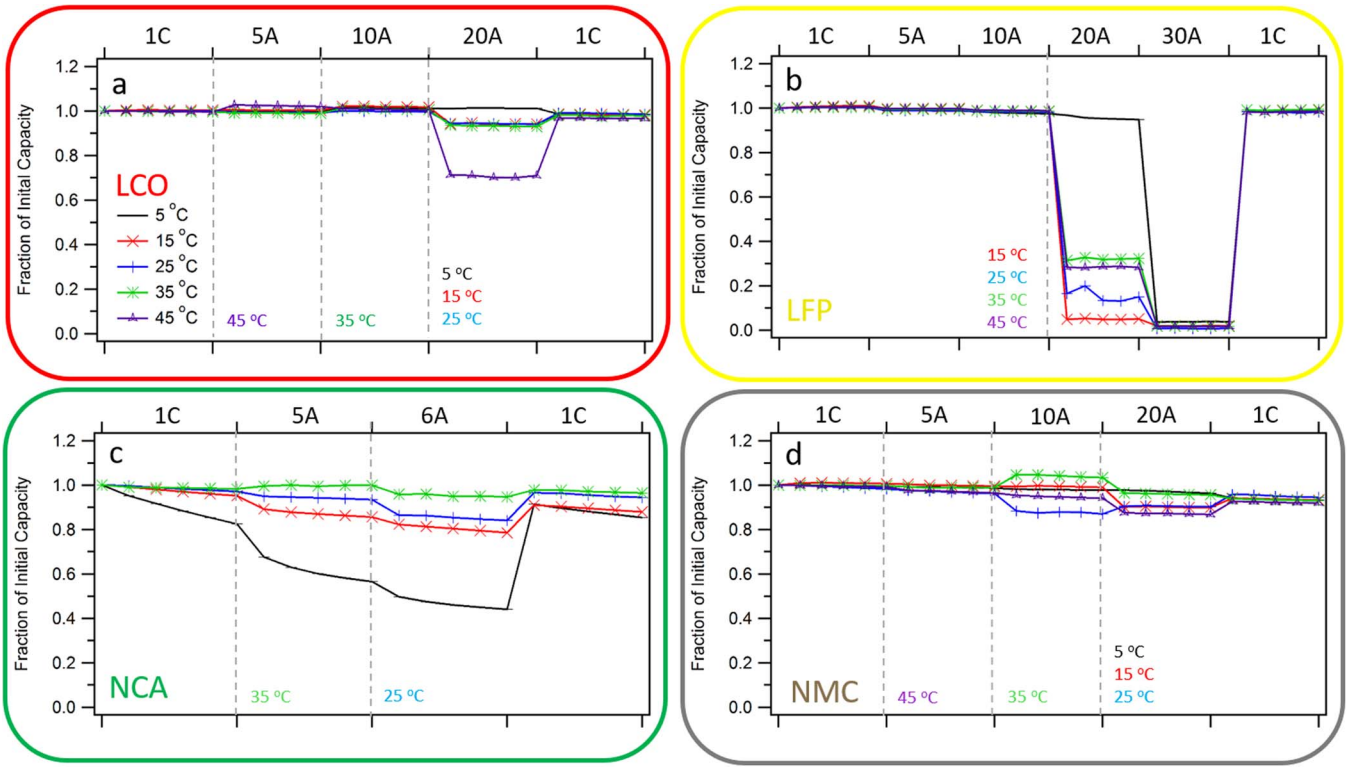


Figure 6. Abuse testing was performed in lieu of segmented discharging. The testing procedure shown in Figure 3 was repeated at 5°C (black dash), 15°C (red x), 25°C (blue cross), 35°C (green star), and 45°C (purple triangle). Recall that the vertical dashed lines indicate the start of segmented discharging for the indicated temperature. Fraction of initial capacity while allowing self-heating and removing temperature abort limits. Data is presented as a function of discharge current (indicated on top of the graph) and environment temperature (indicated by the color of the line) for each cell tested: (a) LCO (red outline, top-left), (b) LFP (yellow outline, top-right), (c) NCA (green outline, bottom-left), and (d) NMC (brown outline, bottom-right).

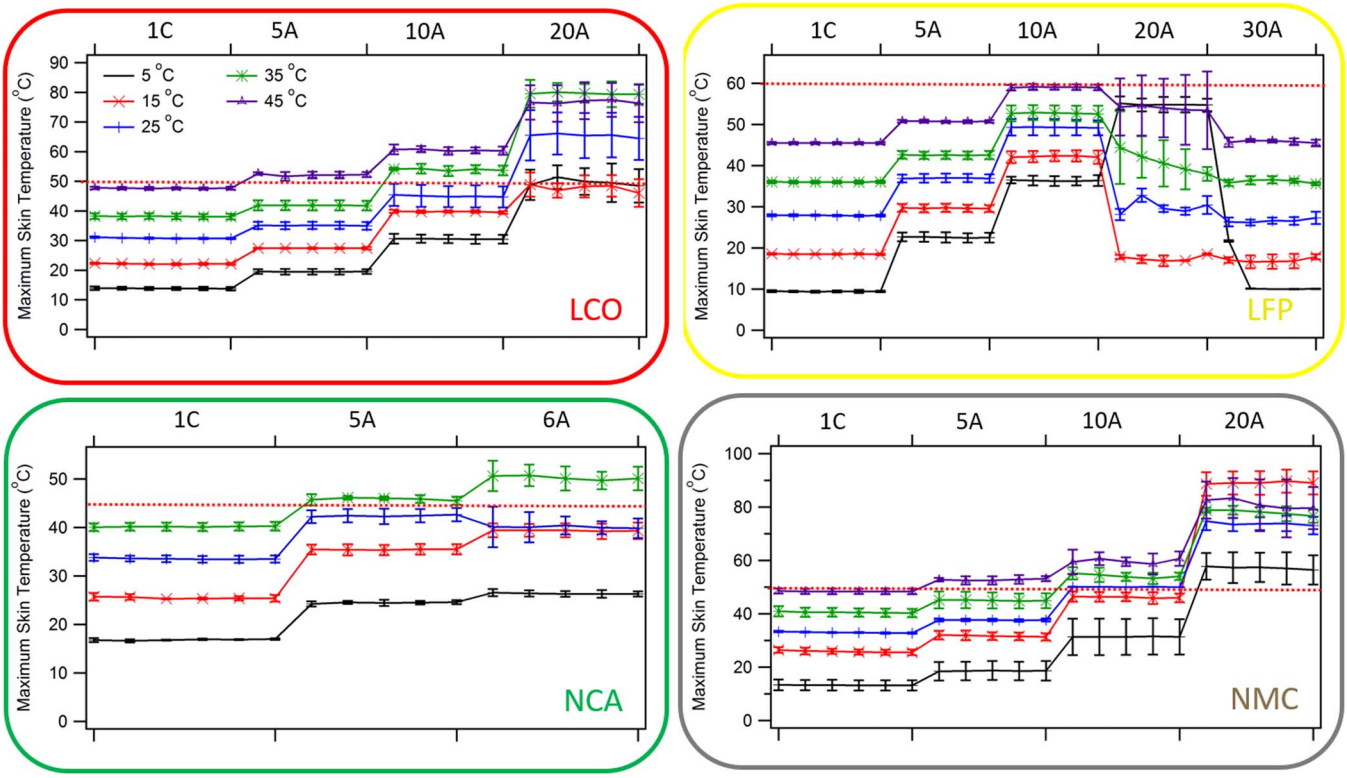


Figure 7. Average and standard deviation of maximum cell skin temperature attained while cycling without temperature abort limits for each cell tested: (a) LCO (red outline, top-left), (b) LFP (yellow outline, top-right), (c) NCA (green outline, bottom-left), and (d) NMC (brown outline, bottom-right). Red dashed lines correspond to the manufacturer specified maximum operating temperature. Anything above the red dashed line is defined as an abuse condition.

above. At higher discharge currents, the discharge capacity diminished significantly. Cells at 5°C and 20A were somewhat shielded from this effect. Increasing to 30 A discharge current resulted in all discharge capacities diminishing to nearly zero. Taking a closer look at the data, the cells reached their end voltage or 2.0 V in less than one second at 30 A discharge. Diminished discharge capacity is likely caused by poor ionic mobility at the high discharge rate.[40–42] Interestingly, segmented discharging reduced this capacity loss. In Figure 5b, the fraction of initial capacity was maintained above 0.5 at temperatures 25°C or above. Likely, returning to OCV during the segmenting allowed the electrode to relax, and greater discharge capacities were realized. All capacity losses shown in Figure 6 were reversible, as illustrated with a post-test return to 25°C and 1C charge/discharge cycling.

As mentioned before, self-heating becomes significant at high discharge currents. To evaluate how hot the cells get, average maximum temperature as a function of cycles is shown in Figure 7. LCO and NMC cells in Figures 7a and 7d, respectively, display significant self-heating. Cells reached nearly 90°C with elevated environment temperature. Self-heating was also evident in Figure 7c. NCA is limited to a discharge current of 6A, which in turn minimizes self-heating. LFP cells, shown in Figure 7b, demonstrate lower self-heating at higher discharge currents. This is easily explained when considering the near-zero discharge capacity observed in Figure 6b. As the discharge capacity and discharge time plummet, the cells are unable to self-heat significantly. At 5°C, when cells still maintain some discharge capacity, self-heating is observed. Although LCO and NMC cells had stable electrochemical performance at high currents, one must carefully control for cell temperature. Taking cells above their allowable operating temperature leaves cells in an unknown, potentially damaged state. These cells may be more likely to experience accelerated aging, especially if the cell is subjected to abuse temperatures repeatedly or for long durations.[43–45] Additionally, exposure to high temperatures may make the cells more prone to go into catastrophic thermal runaway resulting in venting, fire, or explosion.[2,8]

Differential capacity analysis.—To gain a deeper understanding of cell behavior at different environment temperatures, differential capacity analysis of the 1C charge/discharge cycle was performed. Differential capacity at 1C rate is too fast to capture all details of the electrochemical reaction but slow enough to show major processes and changes. Figure 8 shows the differential capacity plots for all cells during the second cycle at 1C charge and discharge. The differential capacity plot for LCO in Figure 8a shows evolution of features as a function of temperature. At 25°C, there are two main charging peaks at 3.54 V and 3.70 V. As the temperature becomes sub-ambient, these peaks shift in both position and height, representing a change in the cell impedance and lithium inventory, respectively.[46] Sub-ambient temperature peak height decrease manifests itself as a capacity loss at 15°C that becomes more pronounced at 5°C. As the temperature increases, both peaks are unchanged but a new peak appears around 3.47 and 3.45 V at 35 and 45°C, respectively. Emerging peaks typically represent a new parasitic reaction.[46] Although parasitic reactions are emerging, they do not appear to alter the cell performance. These reactions may have a more profound impact with prolonged cycling at higher temperatures. In Figure 8b, LFP has three peaks during charging at 25°C: 3.28 V, 3.37 V, and 3.40 V. LFP experiences peak position shifts with temperature, representing impedance changes. The cell is cycled over the entire SOC range, so changes in impedance do not alter the observed capacity. This corresponds well to the stable discharge capacity over all temperatures under 1C charge/discharge rate. NCA displays two major charging peaks in Figure 8c; at 25°C, the peaks are centered around 3.63 V and 3.80 V. Distinct from all other cells, NCA demonstrates a tremendous change in differential capacity at sub-ambient temperatures. At 15°C, a new peak emerges in the high voltage region of the NCA differential capacity plot. Upon cooling further to 5°C, the only major peak is this new high-voltage peak. It is well known that low temperatures can cause lithium plating to occur as the cell reaches 100% SOC.[47] It is likely that lithium plating is beginning to occur here. This matches Figure 4c and Figure 5c as

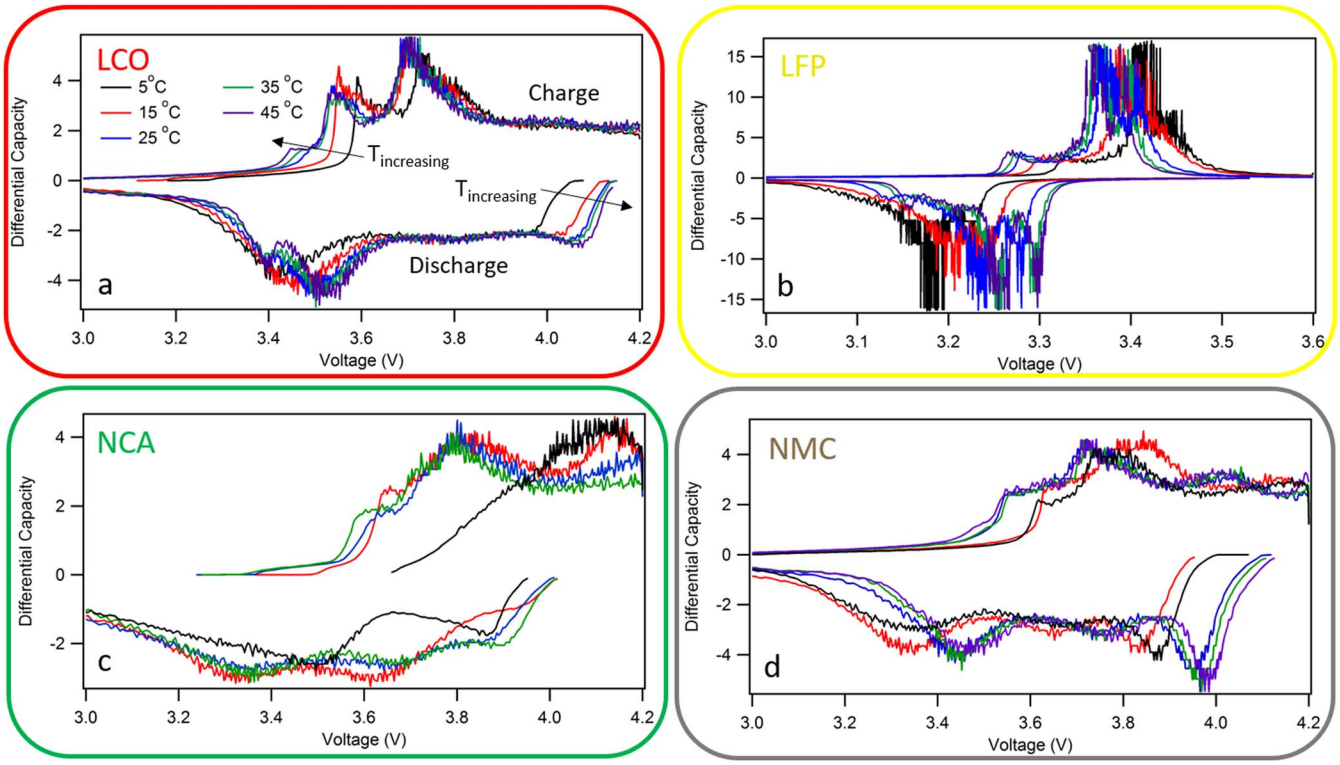


Figure 8. Second cycle differential capacity plots of: (a) LCO (red outline), (b) LFP (yellow outline), (c) NCA (green outline), and (d) NMC (brown outline) at each temperature. Differential capacity was found from the first full 1C charge/ 1C discharge cycle. Curves shifted to lower voltage on charge and higher voltage on discharge.

discharge capacity diminishes at 15°C and 5°C within NCA cells and does not fully return when cycled at 25°C and 1C. Finally, NMC has three main peaks upon charging at 25°C: 3.56 V, 3.72 V, and 4.03 V. Temperature increase does not affect the differential capacity plots. Sub-ambient temperatures shift peak position and height. Peak height loss, corresponding to lithium inventory loss, is most pronounced at 5°C. In Figure 5d, this also represents a loss in discharge capacity compared to higher temperatures. Overall, differential capacity plots provide insights into what chemical changes are responsible for discharge capacity losses displayed in Figures 4 through 6.

Electrochemical impedance spectroscopy.—Electrochemical impedance spectroscopy (EIS) is a powerful tool to gain understanding of internal electrochemical processes. As such EIS was performed on three examples of each cell in two different conditions. EIS was collected after the cells had been cycled at various temperatures. This data was collected on cells at 0% SOC and 25°C, representing irreversible changes within the cell from the test profile. EIS was also collected on fresh cells at 0% SOC and various temperatures. This is meant to ascertain the reversible changes within the cell from environment temperature. Using the equivalent circuit in Figure 9, impedance data was fit. The simple equivalent circuit is a modified Randles circuit which contains a component for the ohmic resistance of the battery (R_{ohm}). This is followed by an *RC* circuit (C_{sei} and R_{sei}) to denote the slow migration of lithium ions through the anode surface film. The final *RC* circuit accounts for the double layer capacitance of electrodes (C_{dl}), a charge transfer resistance (R_{ct}), and diffusion impedance of the anode and cathode (Z_w). Representative EIS data and a corresponding fit is shown in Figure 9. Key fitting results are summarized in Figures 10 and 11.

Reversible changes on pristine, uncycled cells as a function of temperature can be visualized in Figure 10. At lower temperatures, lithium ions have decreased kinetic energy, causing slower movement through the SEI.[45,48] This causes higher R_{sei} and corresponding decreased C_{sei}, Figures 10a and 10b. LFP appears to be an outlier in this trend, where the C_{sei} is about an order of magnitude larger than the other chemistries. This is at odds with the low R_{sei} seen in Figure 10b at 5°C. A possible explanation is that LFP cells are engineered to operate at low temperatures (to −30°C), a higher C_{sei} may be by design to prevent capacity loss. The same kinetic effect results in trends observed in C_{dl} and R_{ct} seen in Figures 10c and 10d. Irreversible changes after cycling were investigated via EIS at 25°C and fitting results are shown in Figure 11. The baseline (before cycling) is shown as the first data point in each graph. Minimal trends evolve in the R_{sei} and R_{ct} examples. If anything, cycling reduced R_{sei} and R_{ct} from the baseline value. LCO and NCA demonstrated increased C_{sei} at elevated temperatures compared to other cells. The C_{dl} was largely consistent, regardless of cycling temperature, and experiences a low C_{dl} value in baseline cells, which increased somewhat after cycling.

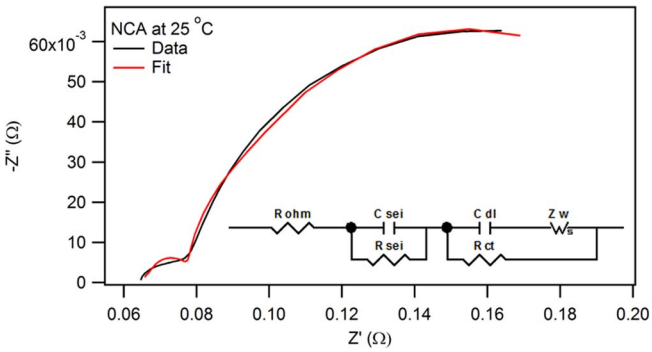


Figure 9. Lithium-ion battery representative EIS data (black line) and fit (red line) and inset equivalent circuit used for fitting in this work. In the inset equivalent circuit from left to right: R_{ohm} represents in ohmic impedance of the entire cell, C_{sei} the capacitance from the SEI layer, R_{sei} the resistance of the SEI layer, C_{dl} the double layer capacitance, R_{ct} the charge transfer resistance, and Z_w the Warburg element of the cell.

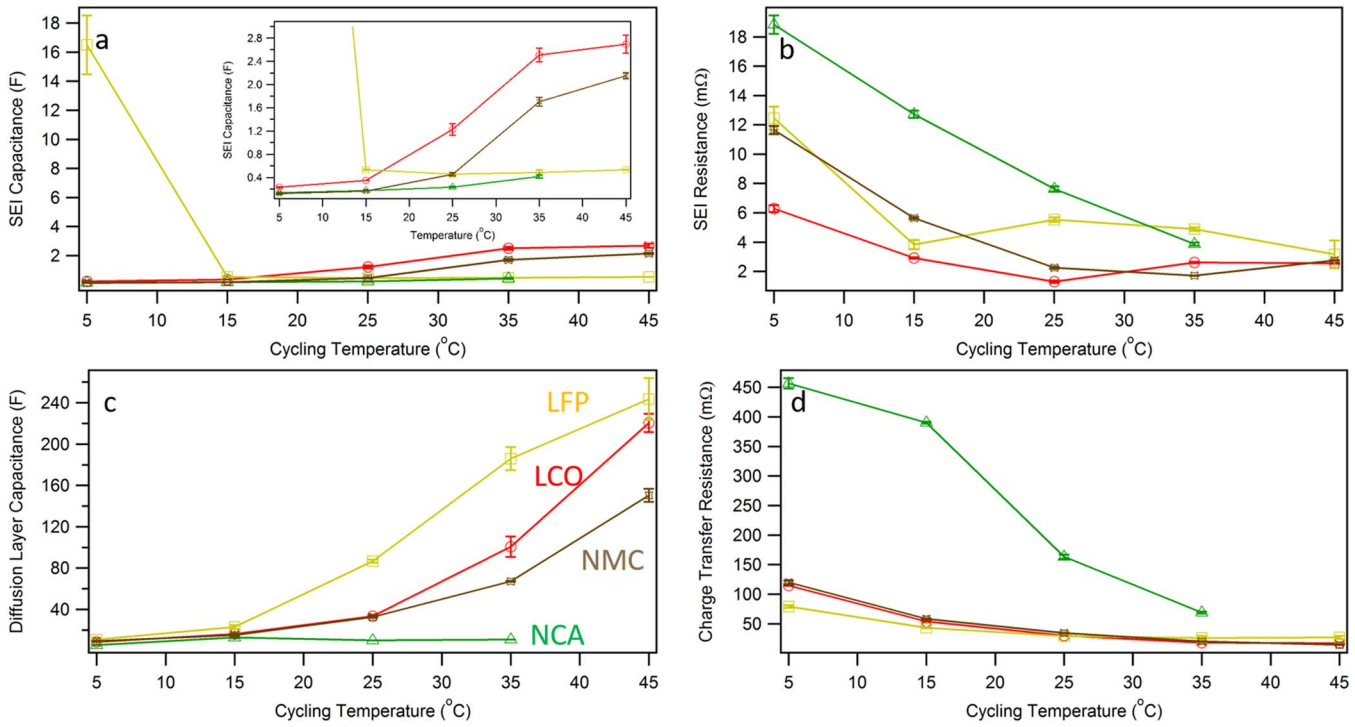


Figure 10. EIS fitting results of pristine batteries: LFP (yellow, open square), LCO (red, open circle), NCA (green, open triangle), and NMC (brown, open star) at various temperatures. (a) SEI capacitance, (b) SEI resistance, (c) diffusion layer capacitance, and (d) charge transfer resistance. ohmic resistance is omitted because all cells had a similar spread.

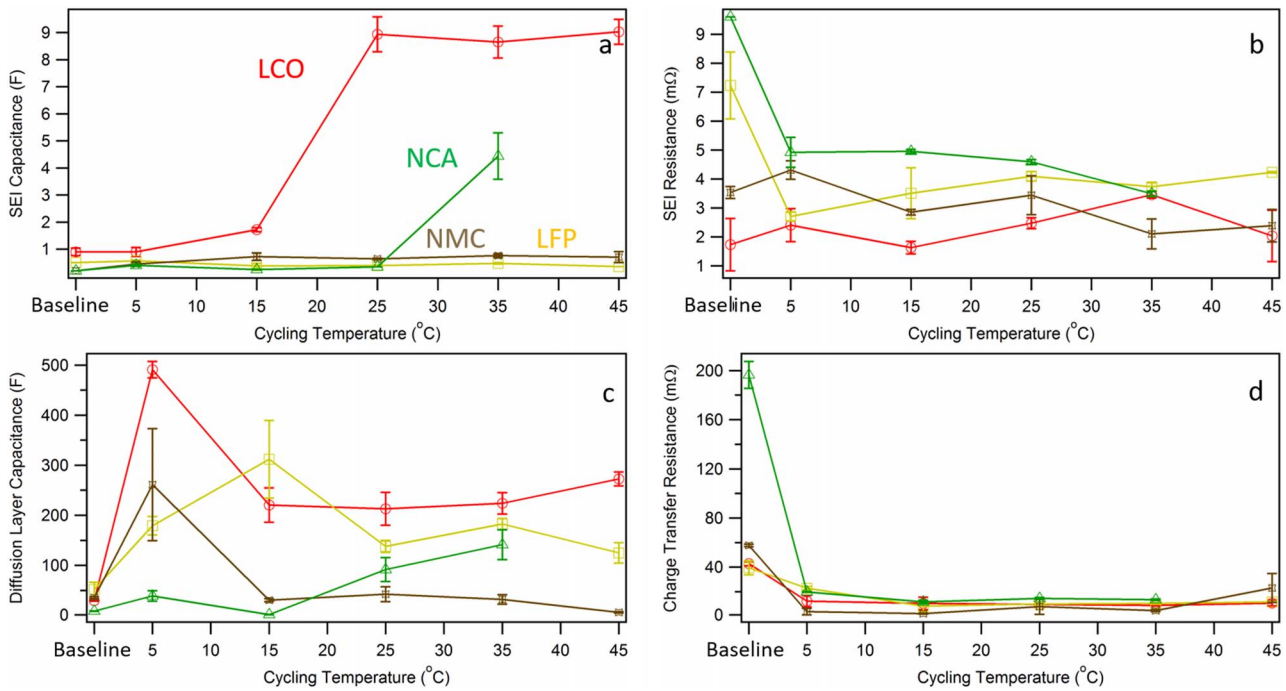


Figure 11. EIS fitting results for batteries: LFP (yellow, open square), LCO (red, open circle), NCA (green, open triangle), and NMC (brown, open star) cycled at various temperatures compared to baseline (first data point in each graph) pristine cells. (a) SEI capacitance, (b) SEI resistance, (c) diffusion layer capacitance, and (d) charge transfer resistance. ohmic resistance is omitted because all cells had a similar spread.

Conclusions

We have analyzed the performance of four commercially relevant lithium-ion cells of various chemistries and manufacturers. A testing methodology consisting of six stages of testing was used to scope overlapping operational conditions of the cells. This testing methodology quantified compounding effects of overlapping operating conditions such as discharge current and environment temperature. Output capacity as a function of operating condition is reported while differential capacity and EIS were used as diagnostic tools to elucidate origins of changes. Initially, energy density, specific energy, and cost per capacity were found for all four cells. Initial capacity testing found that all

four cells performed as expected from manufacturer data sheets at 1C charging and discharging rates. Notable deviations as a function of cell chemistry in these parameters are immediately apparent although consumers must be cautious to consider all application requirements. For example, cycle/calendar life and thermal stability were outside the scope of this work, although they are important parameters to consider when choosing a cell.

We found that electrochemical performance varied greatly with cell chemistry, environment temperature, and discharge current. It is apparent that cells have a desirable operating window wherein operating conditions must be carefully paired with a desired application. We observed that LFP cells displayed temperature-independent performance up through 10 A discharge currents. Operating at high discharge currents led to rapid decrease in output capacity. If maintained at 10 A or below, self-heating was also managed. LCO, NCA, and NMC operated best at 15°C in terms of performance across discharge currents while simultaneously minimizing self-heating. One must still be careful at high currents, even at 15°C environment temperature, to avoid unintended abuse conditions. In NCA, operation across high discharge currents at 15°C aged cells, evidenced by the approximately 10% capacity loss when cycled at 1C and 25°C post-test. Perhaps a lower discharge current would mitigate this aging. Through differential capacity testing, lasting capacity losses experienced by NCA cells at 5 and 15°C can be explained by the appearance of a new, high voltage peak. This new peak likely results from the emergence lithium plating near 100% SOC. In Figures 10b and 10d, SEI and charge transfer resistance values for NCA were also quite high compared to other cells. EIS fitting after cycling failed to reveal additional insights.

Overall, we found that lithium-ion cells have varying responses to operating conditions. Cells are highly application-specific and more information must be available to all consumers for making an informed decision. Without detailed performance data, safety and reliability of commercial cells under specific conditions is not fully understood. For example, compounding operating conditions may elicit unintended abuse conditions while the cells are operated well within the manufacturer's operating range. We showed that high current discharging results in enough self-heating to quickly reach the cutoff temperature. Depending on environment condition, this unintended abuse condition may be mild (< 1°C over limit) to extreme (> 35°C over limit). We also found that unintended abuse conditions can lead to faster cell aging. This could even result in catastrophic thermal runaway and total loss of the ESS. For example, after cycling at 15°C, we found that NCA cells experienced an 11.6% irreversible capacity loss as shown in Table II. It is easy to envision how even small losses quantified in this work can be extrapolated to rapid cell death. However, at sub-ambient temperatures LFP demonstrated an increase in capacity after cycling, up to 3.5% at 15°C, consistent with what is already well known.[36–38] Although the cost per capacity was highest for LFP cells, it does not consider useable lifetime. For this reason alone, further work is needed to continue to extend the scope of this study to other relevant chemistries and cell formats and to extend to quantifying usable lifetime.

Acknowledgments

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